

Emanuel Camacho

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SKILLS

Technical Skills: CAD (Siemens NX, SolidWorks, Solid Edge); CFD (Siemens STAR-CCM+); MATLAB, Simulink, Simscape; Python, C; LaTeX.

Engineering Competencies: Turbomachinery Design, Aerodynamic Analysis, Computational Fluid Dynamics, C-UAS System Architecture, Control Systems (ISMC), Systems Engineering.

Languages: Portuguese (Native), English (C2 - Proficient), Spanish (C1 - Advanced), German (A2 - Elementary).

EXPERIENCE

Aerotéc - BLUE

January 2026 – Present

Design Team Member – Compressor Section

Lisbon, Portugal

- Realized the aerodynamic design and optimization of centrifugal impeller and diffuser components for a small-scale turbojet engine within a 15-member team.
- Built custom MATLAB thermodynamic and mean-line flow scripts to calculate 3D blade geometries, flow paths, and stage pressure ratios.
- Created the entire design team framework for automated parametric CAD generation by interfacing MATLAB scripts with Siemens NX, compressing iterative design cycles into a single button click and 30 seconds processing.
- Executed preliminary 3D CFD simulations in Siemens STAR-CCM+ to analyze boundary layer separation, optimize blade incidence angles and verify theoretically expected pressure ratios.

ANAC – Autoridade Nacional da Aviação Civil

August 2024 – September 2024

Project Intern

Lisbon, Portugal

- Developed aerodynamic failure prediction models for commercial UAS platforms, calculating ballistic trajectories and ground impact zones following critical system failures.
- Designed and implemented an interactive HTML-based simulation tool with an intuitive UI, enabling certification engineers to assess UAS safety parameters.

PROJECT

Instituto Superior Técnico & Portuguese Air Force

September 2025 – November 2026

Master's Thesis | Laser-based Interception of Unmanned Aerial Systems

Lisbon, Portugal

- Defense-Funded Research:** Developed a low-power counter-UAV (C-UAS) prototype in collaboration with and funded by the Portuguese Air Force (*Força Aérea Portuguesa*) and ISR (Institute for Systems and Robotics) to neutralize drone optical sensors.
- Control & Targeting Systems:** Designed and implemented a genetic algorithm optimized Integral Sliding Mode Control (ISMC) scheme on a precision 2-axis gimbal to achieve dynamic visual tracking and optical sensor neutralization.
- Full-System Simulation:** Built an end-to-end Simulink framework integrating high-fidelity gimbal dynamics, visual detection pipelines, sensor noise models, and target trajectory physics prior to hardware deployment.
- Avionics & Hardware Integration:** Prototyped a lightweight C-UAS platform using custom 3D-printed mechanics, embedded computing using a Pixhawk controller, servo motor actuators, and dedicated power regulation architectures.
- Experimental Validation:** Conducted physical field tests to validate optical tracking and neutralization performance, establishing baseline metrics for mobile and drone-interceptor deployments.

CERTIFICATES

Cadflow / Siemens Digital Industries Software

September 2026

STAR-CCM+ CFD Training Certification

Lisbon, Portugal

- Certified training in computational fluid dynamics (CFD) simulation using Siemens STAR-CCM+.

Volkshochschule Lernportal

August 2026

German Language Proficiency (A2)

Online

- Certificate of completion for level A2 German language coursework.

Technical University of Munich

August 2025

German Language Proficiency (A1.1)

Munich, Germany

- Certificate of completion for level A1.1 German language coursework.

EDUCATION

Instituto Superior Técnico

Expected Nov 2026

Master of Science – MSc, Aerospace Engineering

Lisbon, Portugal

- Specialization in Aerodynamics and Aircraft Propulsion Systems.

Technical University of Munich

August 2025

Erasmus Semester – MSc, Aerospace Engineering

Munich, Germany

- Coursework focused on aircraft propulsion and overall aircraft system integration.

Instituto Superior Técnico

July 2024

Bachelor of Science, Aerospace Engineering

Lisbon, Portugal

- Relevant Coursework: Fluid Dynamics, Flight Mechanics, Flight Control.